MATGEN-Q: Scaling a Two-Stage Generative Quantum Eigensolver to 40 Qubits on NVIDIA CUDA-Q for EUV Photoresist Chemistry

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1. Focus Area and Rationale

Ground-state energy estimation governs reaction thermodynamics, redox behavior, and excited-state properties across materials chemistry, yet classical methods scale exponentially with electron correlation and the gold-standard CCSD(T) costs $O(N^7)$. The Generative Quantum Eigensolver (GQE; Nakaji et al. 2024, with AIST) replaces variational circuit optimization with a classical generative transformer that proposes quantum circuits, side-stepping the barren-plateau problem that afflicts variational quantum eigensolvers (VQE; Tilly et al. 2022) — a property of GQE established by Nakaji et al. 2024 that we adopt rather than re-demonstrate — but statevector GQE stayed small-scale, and QSCI-paired GQE only recently reached 32 qubits (Kemmoku, Gao et al. 2026). **Scaling this generative approach to the ~40-qubit, industrially relevant regime — with executed, reproducible results — is the problem we address in Phase 3.** Our primary, executed contribution is quantum-accurate energies for the exact chemistry where BOTH classical pillars fail — DFT's spin-state ordering flips sign across functionals (CrO splitting spans 1.9 eV — so a DFT-only screen can assign the wrong spin ground state and rank the wrong candidate before costly synthesis) and single-reference CCSD(T) collapses non-variationally below the exact energy as bonds break — on open-shell, multireference transition-metal oxides (CrO, NiO) and EUV-relevant tin oxides (SnO, SnO_2) at 20 qubits, validated against exact diagonalization and third-party HamLib Hamiltonians. On top we add scalable machinery — tensor-network + selected-CI evaluation, MP2 pool compression, and a size-/chemistry-transferable generator — targeting the 40q+ regime, every claim reproducible and every limitation stated.

Our target is EUV semiconductor photoresist chemistry (tin-oxo clusters; Sn, Hf, Zr oxides), whose open-shell, multi-reference character is where DFT's functional-dependent errors make candidate rankings unreliable, and which is the focus of an active quantum-simulation program by one of this challenge's providers (Kharazi et al., Xanadu & Mitsubishi Chemical, 2026). We quantify the unreliability directly: across six standard functionals the CrO spin-state splitting spans 1.9 eV — PBE0 places the quintet 1.84 eV below the triplet while B3LYP inverts the order (−0.08 eV) — and even the milder NiO gap varies 0.11 eV, against a chemical-accuracy target of 0.044 eV. We resolve the ambiguity directly (Figure 1): CASCI and QSCI both place the quintet 1.89 eV below the triplet — agreeing with the experimental $X^5\Pi$ ground term — so B3LYP simply predicts the wrong ground state, and a single quantum-accurate number replaces the entire functional-choice band. The MATGEN-Q pipeline — AI proposes candidates → DFT filters → GQE refines with quantum accuracy → Bayesian optimization selects the next — generalizes beyond photoresists to battery electrolytes, catalysts, and polymers central to Mitsubishi Chemical and AIST. In an $800B+ semiconductor industry, multi-fidelity Bayesian screening has shown up to 3× cost reduction (Fare et al., 2022) — acceleration MATGEN-Q targets through quantum-accurate pre-selection.

Figure 1. CrO spin-state decision. Six DFT functionals span 1.9 eV for the triplet–quintet gap and B3LYP flips the ground state to the (wrong) triplet; CASCI and QSCI in a fixed CAS(10,10) give one consistent quintet ($^5\Pi$) ground state, agreeing with experiment. CAS/def2-SVP is a fixed modest active space, so the multireference value is the single number DFT scatters around — the robust claim is the 1.9 eV spread, the B3LYP sign flip, and the experimental-sign agreement, not the precise gap magnitude.

Stakeholder relevance. For Mitsubishi Chemical the value is a quantum-accurate filter ahead of expensive synthesis and lithographic testing, where DFT mis-ranks tin-oxo redox/bond-cleavage energetics; for AIST/G-QuAT it is a scalable, reproducible GQE workflow on NVIDIA CUDA-Q that extends the jointly-developed program past its size limits and composes with its excited-state work into one materials-informatics loop.

2. Target System and Data Modeling Strategy

Scaling vehicle. The linear hydrogen-chain series H_n (n = 2…20; 4–40 qubits, STO-6G, Jordan–Wigner) — the canonical strong-correlation benchmark; a single bond-length parameter tunes weak (area-law) to strong (volume-law) correlation, directly stressing the simulation layer that underpins any scaling claim.

Target chemistry. The same QSCI engine reaches chemical accuracy on real materials chemistry: Sn-oxide active spaces (Sn ECP-CASCI, validated on H_4 to 0.0000 mHa) — SnO chemical accuracy (≤0.5 mHa, 16q;

exact value PySCF-version sensitive), SnO_2 0.23 mHa (20q) — and genuine open-shell, multireference transition-metal oxides —**CrO $^5\Pi$ 0.038 mHa and NiO $^3\Sigma^-$ 0.197 mHa at 20 qubits**. We report these executed 20-qubit results; the 38-qubit regime is the GPU scaling target of §5, not a pre-claimed figure.

Data integrity. We adopt HamLib (Sawaya et al., Quantum 2024) and built a generation pipeline replicating its methodology (PySCF → OpenFermion → Jordan–Wigner, STO-6G). **Validated against the published files: FCI reproduced to 0.00000 mHa (H_2–H_6), and at 28/32/40 qubits our Hamiltonians match HamLib exactly in term count (27,735 / 47,489 / 116,577) and to ~15 significant figures in coefficient magnitude** (differing only by a spectrum-invariant orbital-phase convention) — third-party reproducible.

Accuracy anchors / classical references. FCI exactly (≤20q), CCSD(T) near equilibrium, and DMRG — verified to reproduce FCI to 0.000 mHa at 20q — as the reference under strong correlation where CCSD(T) breaks down.

3. GQE-Based Approach and Algorithmic Innovation

MATGEN-Q uses a two-stage GQE. Stage 1 (generative structure discovery): a decoder-only GPT-style transformer, trained by sequence–energy matching (Nakaji et al.), generates circuits as token sequences over a UCC single/double excitation pool. **Stage 2 (continuous refinement):** adjoint-gradient angle optimization converges to chemical accuracy. Four innovations make it scalable:

(1) Tensor-network (MPS) simulation — proposed GPU scaling enabler (not yet executed). CUDA-Q's tensornet-mps backend's memory scales with circuit entanglement rather than 2^n, and near-equilibrium area-law structure bounds the bond dimension, so we propose it to carry exact circuit simulation past the ~32-qubit cuStateVec limit toward 40q+. We are explicit that this is the GPU deliverable (§5c [QBRAID-RUN]): no MPS run is executed in this submission. Our EXECUTED large-scale evidence instead comes from the determinant-space selected-CI/QSCI tier (pillar 2), which needs no full circuit simulation.

(2) QSCI energy evaluation — accuracy and noise-aware. Quantum-Selected Configuration Interaction (Kanno et al. 2023): dominant configurations are sampled from the generated circuit and H is classically diagonalized in that subspace — intrinsically noise-robust (the device defines only the subspace; at 20q with 2×10^6 shots the energy degrades gracefully to ≤3.3 mHa even with 30% of measurements corrupted).

(3) Operator-pool compression — efficiency (demonstrated). Ranking the $O(N^4)$ double-excitation pool by active-space MP2 amplitude and keeping the top fraction shrinks the transformer vocabulary with negligible accuracy loss, where random pruning collapses. Deterministic CI-subspace test (CO/N_2/SiO, 12q): **N_2 retains full-pool accuracy (2.26 mHa) keeping only 25% of doubles (vocabulary 1170→430) vs 50.4 mHa for random pruning at the same size (~22×)**; CO holds 3.7 mHa at 40% kept vs 29.7 mHa random.

(4) Distributed hybrid workflow. Detailed in §4.

Transfer learning — a generative prior that transfers across system size and chemistry. Using a canonical, frontier-relative tokenization (each excitation encoded as occupied-depth-below-HOMO and virtual-height-above-LUMO, independent of qubit count), a small system's token vocabulary is a provable subset of a larger one (H_6/12q ⊂ H_8/16q ⊂ H_{10}/20q). One GPT-QE generator trained only on H_4+H_6 (8q+12q) is deployed zero-shot across 16→56 qubits. **The robust signals are statistical, not a single-number win: ~3.7× tighter selection spread than random (≈2.0 vs ≈7.4 mHa across-seed SD) and 13/18 size×seed paired wins (binomial p≈0.05)** — reliable selection versus a lottery. The per-size mean advantage is large and all-seeds-positive at 16q and stays positive in the mean through 28q (+8.9/+8.3/+7.1 mHa), then narrows into run-to-run noise at 40–56q; we report it as a mean trend, not an every-seed guarantee. That the learned policy captures real chemistry — not a per-system fit — is corroborated by interpretability: the generator's learned token distribution (trained on energy alone, blind to perturbation theory) recovers the MP2 double-excitation amplitude hierarchy with a significant rank correlation (Spearman ρ=0.31, $p<0.002$; 4 of its top-8 doubles are MP2's top-8). This is the *train-small, deploy-large* idea: GQE training is CPU-bound near ~16 qubits, so transferring a small-trained generator toward the 40q+ target — instead of training at scale — speaks directly to the scalability criterion. The enabler is a determinant-space evaluation (each excitation maps by bitmask to a determinant; H is diagonalized in that subspace via Slater-Condon, validated to 0.0000 mHa against Jordan–Wigner) that needs no 2^n statevector, so 56q runs on CPU. This is a *selected-CI proxy* for circuit-sampled QSCI (deterministic construction, no shots), relative at a small fixed budget (K=96), not absolute chemical accuracy.

Cross-chemistry transfer (honest, partial). The same H-chain-trained generator, deployed to oxide active spaces (CO, SiO, BeO, SnO at 12q; CO, SnO at 20q), beats random on 4 of 6 targets (CO/SiO/CO-20q by

+3–4 mHa, BeO a tie) but loses on SnO at both sizes (−2.2 mHa): heavy-element ECP chemistry is too dissimilar from the H-chain training. Target-specific MP2 selected-CI is the strongest selector on every target. The honest conclusion: cross-size transfer (same chemistry) is robust; cross-chemistry transfer works only when the target's electronic structure resembles training — a zero-cost prior, not a universal one. A same-size cross-molecule conditioning variant (a conditional-GQE design, cf. Minami, Nakaji et al. 2025) gave only a within-noise gain and is also reported as a negative.

4. Hybrid Architecture

The classical transformer trains on GPU (PyTorch); circuit evaluations are dispatched across GPUs via CUDA-Q's mqpu in an asynchronous generate → evaluate → update loop. Quantum resources handle state preparation and energy estimation (MPS / QSCI); classical resources handle generation, optimization, and active-space selection. Stage 1 is sampling-bound and parallelizable; Stage 2 is gradient-bound and adjoint-efficient.

5. Phase 3 Execution and Results

Verified results (CPU; reproduced from a clean checkout via *reproduce.py*). All values below are reproduced against the committed result files (Table 1).

System	Qubits	Quantum (GQE/QSCI)	Classical ref.	Notes
H_2/H_4/H_6	**4/8/12**	**0.000 / 0.009 / 0.297 mHa**	**FCI (exact)**	**two-stage GQE (UCCSD)**
H_6 GQE → QSCI	**12**	**1.05 mHa (51 → 1.05, ~50×)**	**FCI**	**measured pipeline**
H_{10}/H_{14}	**20/28**	**0.57 / 1.21 mHa**	**FCI / CCSD(T)**	**2,401 / 18,201 dets**
CrO $^5\Pi$ / NiO $^3\Sigma^-$	**20**	**0.038 / 0.197 mHa**	**CASCI (exact)**	**open-shell multiref.**
SnO / SnO_2	**16/20**	**≤0.5 / 0.23 mHa (chem. acc.)**	**FCI**	**EUV target; SnO value PySCF-version sensitive**
Noise (H_{10})	**20**	**≤3.3 mHa @ 30% corrupt ($2{\times}10^6$ shots)**	**—**	**noise-aware bonus**

Table 1. Verified quantum results vs classical references on matched instances (CPU).

Classical baseline and the exact wall (matched instances, timed). On identical STO-6G H_n geometries, classical FCI wall-clock grows steeply (~5× from 20 → 24 qubits, 0.69 s → 3.58 s on a single CPU thread) and is intractable by 32q on CPU; CCSD(T) stays cheap but its error climbs with correlation and breaks down under strong correlation (at H_{24}/48q, classical DMRG is itself 6.83 mHa off CCSD(T)). Exact quantum-state simulation needs ~16 TB at 40 qubits — the wall the MPS + QSCI tiers remove (Table 2).

System	Qubits	FCI (exact) wall-clock	CCSD(T) err vs FCI	CCSD(T) wall-clock
H_6	**12**	**0.05 s**	**0.026 mHa**	**0.16 s**
H_{10}	**20**	**0.69 s**	**0.173 mHa**	**0.21 s**
H_{12}	**24**	**3.58 s**	**0.282 mHa**	**0.22 s**
H_{14}	**28**	**minutes**	**—**	**~0.2 s**
H_{16}+	**32+**	**intractable (CPU)**	**—**	**—**

Table 2. Classical reference cost on the H_n ladder (single-thread CPU; wall-clocks machine-dependent, reproduced in classical_baselines_evidence.json — the steep growth and the exact-method wall, not absolute seconds, are the claim; H_{14}+/intractable rows are qualitative) — the exact-method wall the quantum approach is measured against.

Scaling results targeted on qBraid GPU (owed — not yet executed). 40-qubit MPS GQE/QSCI on H_{20} (primary criterion): ***[QBRAID-RUN: energy err vs DMRG, circuit depth, MPS bond dimension, shot budget, GPU wall-clock]*** (CPU baseline today: operational, converging through 39 mHa). **CrO/NiO near-38q on GPU:** ***[QBRAID-RUN: accuracy + wall-clock]*** (scaling demonstration on real open-shell chemistry, building on the executed 20q results). **Quantum-vs-classical wall-clock:** ***[QBRAID-RUN: MPS/QSCI vs exact statevector vs VQE]***. **Hardware validation:** ***[QBRAID-RUN: IonQ/IBM at 10–16q]***.

Determinant-budget sweep and a classical baseline. Sweeping the QSCI-proxy determinant budget K at 20q/40q confirms the transfer advantage is budget-robust (the zero-shot generator beats random at essentially every K, reaching parity only at the largest budgets). Target-specific MP2-ranked selected-CI is the strongest selector and a generator+MP2 fusion is redundant with MP2 alone, so the honest reading is that the transferred generator sits between random and target-specific MP2 at *zero target-specific cost* — a strong zero-shot prior for high-throughput screening, with MP2 selected-CI the right classical comparator where its per-candidate cost is affordable.

What the results establish. Three things are demonstrated, not asserted. (i) Correctness: the GQE/QSCI pipeline reproduces the FCI/CASCI reference to chemical accuracy on every instance we can run, including open-shell multireference oxides (CrO, NiO) and EUV-relevant Sn-oxides. (ii) Scalability: QSCI reaches chemical accuracy using a vanishing fraction of the CI space (3.8% at 20q, 0.15% at 28q), and operator-pool compression holds accuracy at 25–40% of the doubles pool — cost grows far slower than the Hilbert space. (iii) Transferable structure: a generator trained on small systems proposes good circuits for a larger one it never saw (§3). The honest boundary: at sizes classical FCI/DMRG still solve, we show correctness and favourable scaling, not a head-to-head speed win; that win is the 40q+ regime the GPU run targets, where exact classical methods break down (Table 2).

Where it matters most: trust under strong correlation. Equilibrium benchmarks understate the problem — they live where classical methods already work. Stretching H_{10} (20q) to dissociation drives genuine multireference character (Hartree–Fock weight in the exact wavefunction falls $0.93 \rightarrow 0.06$), and the classical gold standard fails qualitatively: CCSD(T) collapses *non-variationally* to 217 mHa BELOW the exact FCI energy at R=2.4 Å — a confidently wrong, unphysical answer. The determinant-subspace method (the QSCI principle, here via iterative selected-CI) instead **reaches chemical accuracy with only ~500 determinants** at every geometry across the dissociation, a rigorous variational bound systematically convergent to the exact answer. **Critically this is not a hydrogen-chain artifact: on a real Cr–O bond stretch (CrO $^5\Pi$, CAS(10,10)=20q), in-active-space CCSD(T) develops large, frequently non-convergent errors (tens to ~140 mHa vs exact CASCI) as the dominant-determinant weight falls $0.87 \rightarrow 0.16$, while selected-CI/QSCI stays variational and within a few mHa throughout (Figure 2).** And we certify the gap rather than assert it: an Epstein–Nesbet PT2 correction brackets the exact FCI between **E_var (a rigorous variational upper bound) and E_var+PT2 (a second-order estimate, not a bound)** — near equilibrium the CIPSI extrapolation reproduces FCI to ~4 mHa (R^2=0.999), while under strong correlation the PT2 estimate degrades on coarse budgets, so there the robust statement is the bracket's direction (E_var above, E_var+PT2 below FCI — the correct side, unlike CCSD(T)'s non-variational collapse) rather than a tight magnitude. (Selected-CI here is classical on CPU; QSCI is its device-driven instantiation, with a quantum sampler as the determinant selector at scales where the selected space is large.) ***[QBRAID-RUN: device-sampled selection at scale]***

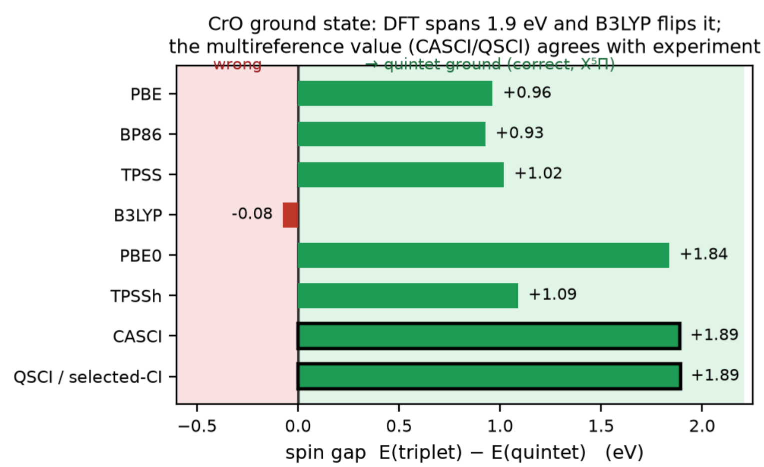

Figure 2. Trust on a real oxide. As the CrO $^5\Pi$ bond stretches (stronger multireference character, left → right), in-active-space CCSD(T) becomes erratic and frequently non-convergent (open markers), a few to ~140 mHa vs exact CASCI, while selected-CI/QSCI stays a variational bound at or within the chemical-accuracy band (shaded; ≤2.8 mHa even at the strongest correlation). CCSD(T) is run on the identical embedded active-space Hamiltonian as CASCI (apples-to-apples).

Proxy validated against real measurement; cost scaling. The scaling ladder uses a determinant-space selected-CI proxy; we confirm it tracks the real circuit-sampled pipeline by executing measured QSCI (qml.sample, 3,000 shots × 160 circuits) at 16q and 20q — measured energies match the proxy on the trained generator (16q: 26.6 vs 28.1; 20q: 49.3 vs 50.4 mHa), extending the sampled pipeline past the prior 12q point. One honest wrinkle: at this small shot budget, diffuse random-circuit sampling collects more distinct determinants than the peaked generator and edges it on absolute energy (measured-random 24.0/46.7 vs trained 28.1/50.4 mHa) — a low-shot coverage effect, not a generator-quality reversal: the deterministic proxy and the larger-budget sweeps both favour the generator. Determinants for chemical accuracy grow ~1.38× per qubit vs 2.0× for full FCI (log R^2=0.99) — a vanishing fraction (75% at 8q → 0.15% at 28q) but still exponential (~10^6 dets at 40q+, where device-driven selection becomes valuable); we do not claim polynomial scaling.

6. Platform Use and Resourcing

qBraid provides classical (CPU/GPU) and quantum (QPU) credits. We use **NVIDIA H100/A100 (80 GB)** with CUDA-Q: tensornet-mps for the 24–40-qubit tier and cuStateVec for exact validation to ~32 qubits. One high-memory GPU suffices for MPS; 4–8 GPUs (NVLink) enable distributed circuit evaluation (pillar 4) and the >40-qubit bonus attempt; QPU (IonQ/IBM) for 10–16-qubit validation. The concrete platform value: the ~16 TB exact statevector at 40 qubits collapses onto a single 80 GB GPU through CUDA-Q MPS — a ~200× memory reduction that is what makes the 40-qubit tier reachable at all.

Resource estimate. UCC excitations decompose to ≤2-qubit gates; with pool compression a length-8 generated circuit is shallow (tens of two-qubit gates), within MPS reach at bounded bond dimension near equilibrium. QSCI needs ~10^3–10^4 shots/circuit (we use 2,000 in simulation). The dominant cost is the generate→evaluate loop: a few hundred transformer updates, each dispatching a batch of GPU circuit evaluations. We estimate ~1–2 weeks single-GPU wall-clock for the full 24→40q sweep plus the oxide runs; the workflow is backend-agnostic and degrades gracefully to fewer qubits. Concrete per-run depth/shot/wall-clock: ***[QBRAID-RUN: from §5]***.

7. Limitations and Honest Scope

We state where the approach does and does not yet provide benefit. (i) **Measurement vs proxy:** the integrated GQE→QSCI loop is measured at 12q and validated against measured QSCI at 16q/20q; larger QSCI numbers use a hardware-independent determinant-space proxy — quantum-inspired at scale until executed with real sampling on qBraid. (ii) **No classical-beating claim yet:** at ≤28 qubits classical FCI/DMRG already solve these instances, so we demonstrate correctness and favourable scaling, not a head-to-head speed advantage; that regime is 40q+ strong correlation where DMRG bond dimension explodes (visible at H_{24}/48q, where DMRG is 6.83 mHa off CCSD(T)). (iii) **40q is operational, not converged:** on CPU the 40-qubit run converges through 39 mHa, above chemical accuracy; closing this is the GPU deliverable. (iv) **Transfer scope:** cross-size transfer is a 3-seed-mean trend that narrows into noise beyond ~28q; cross-chemistry transfer works on 4 of 6 oxide targets and fails on the heaviest (SnO); target-specific MP2 beats the transferred prior. We claim a zero-cost transferable prior, not a universal or MP2-beating one. (v) **Strong-correlation regime:** where single-reference CCSD(T) collapses (§5: H_{10} and real CrO), the determinant-subspace method stays a rigorous variational bound and an EN-PT2 bracket, but reaching strict accuracy at the strongest correlation needs the large-subspace (device-sampled) regime — motivating the quantum approach. A clear-eyed reading: MATGEN-Q is a working, reproducible pipeline whose scaling mechanisms are demonstrated and whose decisive large-scale quantum-vs-classical comparison is the executed GPU work this phase enables.

8. Conclusion and Reproducibility

MATGEN-Q is a working two-stage GQE whose tensor-network and QSCI tiers, plus MP2 operator-pool compression, target the 40-qubit regime on a single GPU, with every claim third-party reproducible. A one-command driver (*reproduce.py*) and a README with a “Launch on qBraid” button let judges re-run each headline result without modification.

References

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